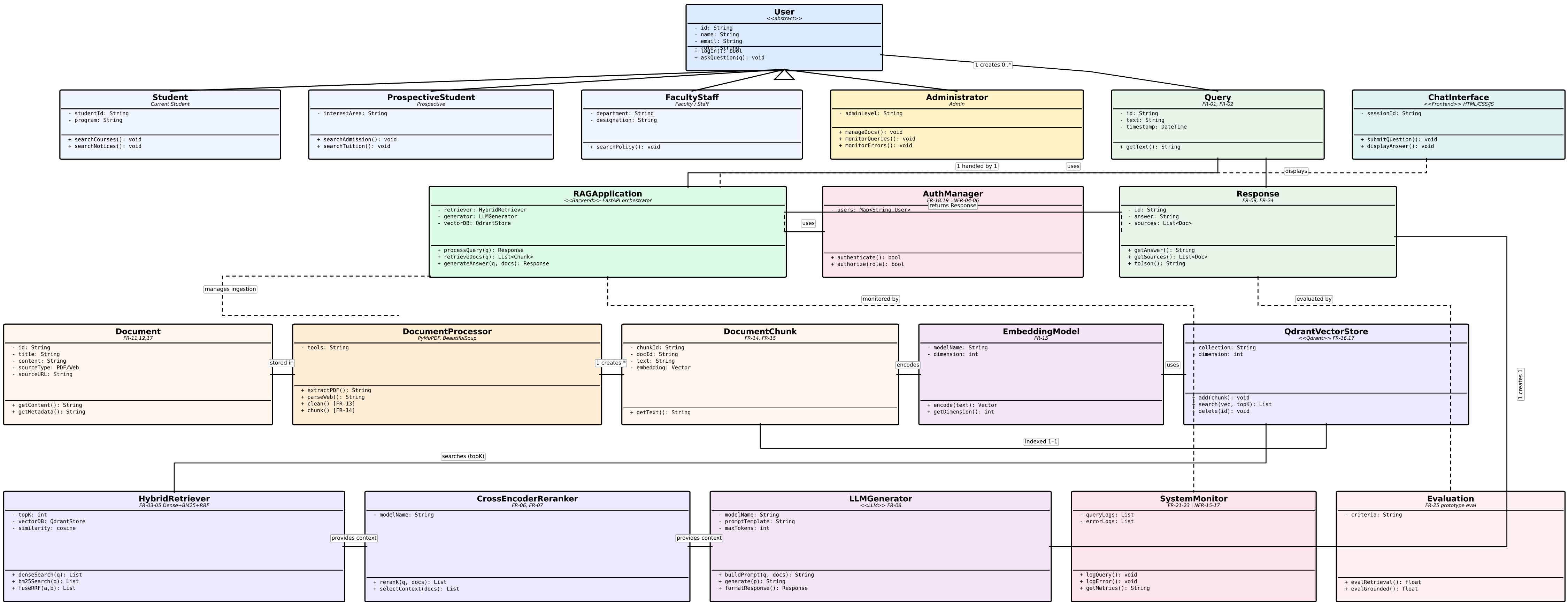


Production-Based RAG System for University Websites — Class Diagram

Fahim | Roll: 24524203101 | Sec-2 Overall Description + System-wide (FR-01-FR-25, NFR-01-NFR-22) | FastAPI, Qdrant, PyMuPDF, BeautifulSoup



Legend + RAG Workflow (Simplified)

— Association (solid, label = role / multiplicity)

--- Dependency (uses / monitored by / evaluated by)

△ Inheritance (is-a: Student, Prospective, Faculty, Admin → User)

- User submits query via ChatInterface → Query → RAGApplication
- RAGApplication → HybridRetriever (Dense + BM25 → RRF) → Rerank → Context
- LLMGenerator builds prompt + generates grounded answer + sources → Response
- DocumentProcessor (PyMuPDF/BS4) → Chunk → Embedding → Qdrant (indexed)
- AuthManager + SystemMonitor + Evaluation are cross-cutting

SRS Coverage (why this is implementation-aligned, not generic)

- Sec-2 (Fahim): 4 user classes + Frontend/Backend + Qdrant / PyMuPDF / BS4 + prototype scope
- FR-01/02/10: Query + ChatInterface + RAGApplication • FR-03~07: HybridRetriever + CrossEncoderReranker
- FR-08/09/24: LLMGenerator → Response (answer + sources) • FR-11~17: Document → Processor → Chunk → Embedding → Qdrant
- FR-18~23 + NFR-04~06,15~17: AuthManager + SystemMonitor • FR-25: Evaluation • NFR-18: separated components

File: class_diagram_fahim_24524203101.png / .pdf • Print: A3 landscape, 200dpi

Note: classdiagram_wasit.png is generic — this diagram maps 1-to-1 to FR/NFR + Sec-2 stack.

How to submit: git checkout -b fahim-class-diagram → add these 2 files → push → Pull Request to main